

Career Paths & Course Catalogue

Strategic Analysis — 2025-2026

Prepared: June 07, 2026

Executive Summary

Metric	Value
Total career paths	10
Total unique courses	45
Shared courses (2+ paths)	5
Patch 1 paths (launch-ready)	5 — AI Engineer, DevOps, Cloud, Data Engineer, Full Stack Java
Patch 2 paths	5 — Frontend, MERN, Data Scientist, Security, Mobile
Avg courses per path	4.5 (down from 5.2 before optimisation)
Estimated Patch 1 gen cost	~22 courses × \$0.03 avg = ~\$0.66 (Haiku model)

Industry Demand Rankings — All 10 Paths

Rankings based on job-posting volume (LinkedIn/Indeed 2025 Q1), YoY salary growth, learner enrolment trends and strategic ROI for the platform.

Rank	Career Path	Difficulty	Duration	Demand	Salary Range (USD)	Patch
1	AI Engineer	Advanced	8 months	★★★★★	\$145–195K	1 ■
2	DevOps Engineer	Advanced	5 months	★★★★★	\$130–175K	1 ■
3	Cloud Engineer	Advanced	6 months	★★★★½	\$135–180K	1 ■
4	Data Engineer	Advanced	6 months	★★★★½	\$130–170K	1 ■
5	Full Stack Java Developer	Advanced	7 months	★★★★	\$120–160K	1 ■
6	Frontend Engineer	Intermediate	4 months	★★★★	\$100–140K	2 ■
7	MERN Stack	Intermediate	5 months	★★★★	\$100–140K	2 ■
8	Data Scientist	Intermediate	6 months	★★★½	\$115–155K	2 ■
9	Web & Cloud Security Engineer	Advanced	6 months	★★★★	\$130–175K	2 ■
10	Mobile Developer	Intermediate	5 months	★★★	\$105–145K	2 ■

Key insight: Patch 1 focuses on infrastructure + AI roles where job-market demand is highest and salary premiums are strongest. Patch 2 covers high-learner-volume tracks where content overlap reduces net production cost

significantly.

Patch 1 — Launch-Priority Career Paths

These 5 paths should be generated and published first. They target the highest-paying, fastest-growing roles and require 22 unique course topics (some shared).

1. AI Engineer ● Advanced · 8 months · \$145–195K

Fastest-growing tech role in 2025–26. LLM-native tooling is mainstream; demand for production AI outpaces supply 3:1.

#	Course	Category	Difficulty
1	Python for AI/ML	AI & Machine Learning	Beginner
2	Large Language Models	AI & Machine Learning	Intermediate
3	Retrieval-Augmented Generation	AI & Machine Learning	Intermediate
4	AI Agents & Agentic Workflows	AI & Machine Learning	Advanced
5	MLOps & Model Deployment	AI & Machine Learning	Advanced

2. DevOps Engineer ● Advanced · 5 months · \$130–175K

Evergreen demand. Platform engineering is the new DevOps title; Kubernetes + Terraform are must-haves for every cloud hire.

#	Course	Category	Difficulty
1	Linux & Shell Scripting	DevOps	Beginner
2	Docker & Containers	DevOps	Beginner
3	Kubernetes	DevOps	Intermediate
4	Terraform & Infrastructure as Code	DevOps	Intermediate
5	CI/CD with GitHub Actions	DevOps	Beginner

3. Cloud Engineer ● Advanced · 6 months · \$135–180K

AWS holds 33 % cloud market share; SAA-C03 remains the #1 entry-level cloud cert by job-listing volume.

#	Course	Category	Difficulty
1	AWS Core Services	Cloud	Beginner
2	AWS Solutions Architect	Cloud	Intermediate
3	Terraform & Infrastructure as Code	DevOps	Intermediate
4	Cloud Security	Cloud	Intermediate
5	Serverless Architecture	Cloud	Intermediate

4. Data Engineer ● Advanced · 6 months · \$130–170K

Data pipelines underpin every AI initiative. dbt + Airflow + Spark is now the standard modern data stack hired by 70 % of data teams.

#	Course	Category	Difficulty
1	SQL Mastery	Data Engineering	Beginner
2	Apache Kafka & Messaging	Data Engineering	Intermediate
3	Apache Spark	Data Engineering	Intermediate

4	Apache Airflow & Orchestration	Data Engineering	Intermediate
5	dbt & Analytics Engineering	Data Engineering	Intermediate

5. Full Stack Java Developer ● Advanced · 7 months · \$120–160K

Java remains the dominant enterprise language. Spring Boot + React is the standard stack at banks, fintechs and large ISVs.

#	Course	Category	Difficulty
1	Java Spring Boot	Backend	Intermediate
2	PostgreSQL Mastery	Backend	Intermediate
3	React.js	Frontend	Beginner
4	Docker & Containers	DevOps	Beginner

Patch 2 — Second-Wave Career Paths

These 5 paths leverage high content-overlap with Patch 1 tracks, cutting net production cost. Frontend, MERN and Mobile share TypeScript and React with Full Stack Java and MERN.

6. Frontend Engineer ● Intermediate · 4 months · \$100–140K

React + Next.js + TypeScript is the dominant frontend trio. Strong overlap with MERN and Full Stack Java paths reduces content cost.

#	Course	Category	Difficulty
1	TypeScript	Frontend	Beginner
2	React.js	Frontend	Beginner
3	Next.js	Frontend	Intermediate
4	Tailwind CSS & Modern CSS	Frontend	Beginner

7. MERN Stack ● Intermediate · 5 months · \$100–140K

Most popular "first full-stack" learning path globally. 3 of 4 courses shared with Frontend/Full Stack Java reduces net cost.

#	Course	Category	Difficulty
1	React.js	Frontend	Beginner
2	Node.js & Express	Backend	Beginner
3	TypeScript	Frontend	Beginner
4	MongoDB	Backend	Beginner

8. Data Scientist ● Intermediate · 6 months · \$115–155K

Strong learner demand but role is evolving toward AI Engineer. Ranked 8th to allow Patch 1 AI Engineer track to launch first.

#	Course	Category	Difficulty
1	Python for AI/ML	AI & Machine Learning	Beginner
2	SQL Mastery	Data Engineering	Beginner
3	Machine Learning Fundamentals	AI & Machine Learning	Beginner
4	Data Science with Python	AI & Machine Learning	Beginner

9. Web & Cloud Security Engineer ● Advanced · 6 months · \$130–175K

Defensive/application security scope only (renamed from Cybersecurity). High industry demand but niche learner base; Patch 2 placement is optimal.

#	Course	Category	Difficulty
1	Linux & Shell Scripting	DevOps	Beginner
2	Web App Security OWASP	Security	Intermediate
3	API Security	Security	Intermediate
4	Cloud Security Fundamentals	Security	Beginner
5	DevSecOps	DevOps	Intermediate

10. Mobile Developer ● Intermediate · 5 months · \$105–145K

Healthy but slower-growth segment. React Native + Flutter dual coverage maximises reach. TypeScript shared with Frontend/MERN paths.

#	Course	Category	Difficulty
1	TypeScript	Frontend	Beginner
2	React Native	Frontend	Intermediate
3	Flutter	Frontend	Intermediate
4	API Design & Best Practices	System Design	Intermediate

Course Sharing Map

Courses appearing in multiple paths are generated once and reused, reducing cost and ensuring consistency. Shared courses are shown in **bold** in the path detail above.

Course	Used in Paths	Count
TypeScript	Frontend, MERN, Mobile	3
React.js	Frontend, MERN, Full Stack Java	3
Docker & Containers	DevOps, Full Stack Java	2
Terraform & Infra as Code	DevOps, Cloud	2
Python for AI/ML	AI Engineer, Data Scientist	2
Linux & Shell Scripting	DevOps, Security	2
SQL Mastery	Data Engineer, Data Scientist	2
Cloud Security	Cloud, Security	2
Next.js	Frontend	1
Tailwind CSS	Frontend	1
Node.js & Express	MERN	1
MongoDB	MERN	1
Spring Boot	Full Stack Java	1
PostgreSQL Mastery	Full Stack Java	1
Kubernetes	DevOps	1
AWS Core / Solutions Arch	Cloud	1
Serverless Architecture	Cloud	1
LLMs	AI Engineer	1
RAG	AI Engineer	1
AI Agents	AI Engineer	1
MLOps	AI Engineer	1
Kafka	Data Engineer	1
Apache Spark	Data Engineer	1
Airflow	Data Engineer	1
dbt	Data Engineer	1
ML Fundamentals	Data Scientist	1
Data Science with Python	Data Scientist	1
Web App Security OWASP	Security	1
API Security	Security	1
DevSecOps	Security	1
React Native	Mobile	1
Flutter	Mobile	1

API Design	Mobile	1
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Full Course Catalogue — 61 Courses

Complete list of all courses in the platform catalogue, organised by category.

#	Course Name	Category	Difficulty
1	Python for AI/ML	AI & ML	Beginner
2	Large Language Models	AI & ML	Intermediate
3	AI Agents & Agentic Workflows	AI & ML	Advanced
4	Retrieval-Augmented Generation	AI & ML	Intermediate
5	PyTorch Deep Learning	AI & ML	Intermediate
6	TensorFlow & Keras	AI & ML	Intermediate
7	Hugging Face Transformers	AI & ML	Intermediate
8	MLOps & Model Deployment	AI & ML	Advanced
9	Data Science with Python	AI & ML	Beginner
10	Machine Learning Fundamentals	AI & ML	Beginner
11	Generative AI for Developers	AI & ML	Intermediate
12	Vector Databases & Embeddings	AI & ML	Intermediate
13	AWS Core Services	Cloud	Beginner
14	AWS Solutions Architect	Cloud	Intermediate
15	AWS Data & Analytics	Cloud	Intermediate
16	Google Cloud Platform	Cloud	Beginner
17	Google Cloud AI/ML	Cloud	Intermediate
18	Microsoft Azure Fundamentals	Cloud	Beginner
19	Azure AI Services	Cloud	Intermediate
20	Cloud Security	Cloud	Intermediate
21	Serverless Architecture	Cloud	Intermediate
22	Docker & Containers	DevOps	Beginner
23	Kubernetes	DevOps	Intermediate
24	Terraform & Infrastructure as Code	DevOps	Intermediate
25	CI/CD with GitHub Actions	DevOps	Beginner
26	Linux & Shell Scripting	DevOps	Beginner
27	Monitoring & Observability	DevOps	Intermediate
28	Site Reliability Engineering	DevOps	Advanced
29	DevSecOps	DevOps	Intermediate
30	Python FastAPI/Django	Backend	Beginner
31	Node.js & Express	Backend	Beginner
32	Java Spring Boot	Backend	Intermediate
33	Go Golang	Backend	Intermediate
34	Rust Programming	Backend	Advanced
35	GraphQL APIs	Backend	Intermediate
36	Microservices Architecture	Backend	Advanced

37	Apache Kafka & Messaging	Backend	Intermediate
38	PostgreSQL Mastery	Backend	Intermediate
39	MongoDB	Backend	Beginner
40	SQL Mastery	Data Engineering	Beginner
41	Apache Spark	Data Engineering	Intermediate
42	dbt & Analytics Engineering	Data Engineering	Intermediate
43	Apache Airflow & Orchestration	Data Engineering	Intermediate
44	Snowflake BigQuery	Data Engineering	Intermediate
45	Real-time Data Streaming	Data Engineering	Advanced
46	Data Warehouse Design	Data Engineering	Intermediate
47	React.js	Frontend	Beginner
48	Next.js	Frontend	Intermediate
49	TypeScript	Frontend	Beginner
50	Vue.js Nuxt.js	Frontend	Intermediate
51	Tailwind CSS & Modern CSS	Frontend	Beginner
52	React Native	Frontend	Intermediate
53	Flutter	Frontend	Intermediate
54	System Design Interview Prep	System Design	Advanced
55	Distributed Systems	System Design	Advanced
56	API Design & Best Practices	System Design	Intermediate
57	Redis & Caching Strategies	System Design	Intermediate
58	Database Scaling & Sharding	System Design	Advanced
59	Web App Security OWASP	Security	Intermediate
60	API Security	Security	Intermediate
61	Cloud Security Fundamentals	Security	Beginner

Difficulty colour legend: **Beginner** **Intermediate** **Advanced**

Generation Cost Estimate

Using Claude Haiku at \$0.80/M input + \$4/M output tokens. Each lesson: ~2,000 input tokens + ~3,500 output tokens = ~\$0.0156/lesson. 35 lessons per course.

Patch	Unique Courses	Lessons	Est. Cost	Notes
Patch 1	22	770	\$12.00	AI, DevOps, Cloud, Data Eng, Java — no prior content
Patch 2	10*	350	\$5.46	*Excludes 13 courses shared with Patch 1
Total	32	1,120	\$17.46	Remaining 29 catalogue courses for future phases

Note: 8 AI/ML courses (indexes 0–11 minus Data Science with Python) have already been generated in a prior run at ~\$4.42. The remaining 22 Patch-1 courses are the priority generation target.